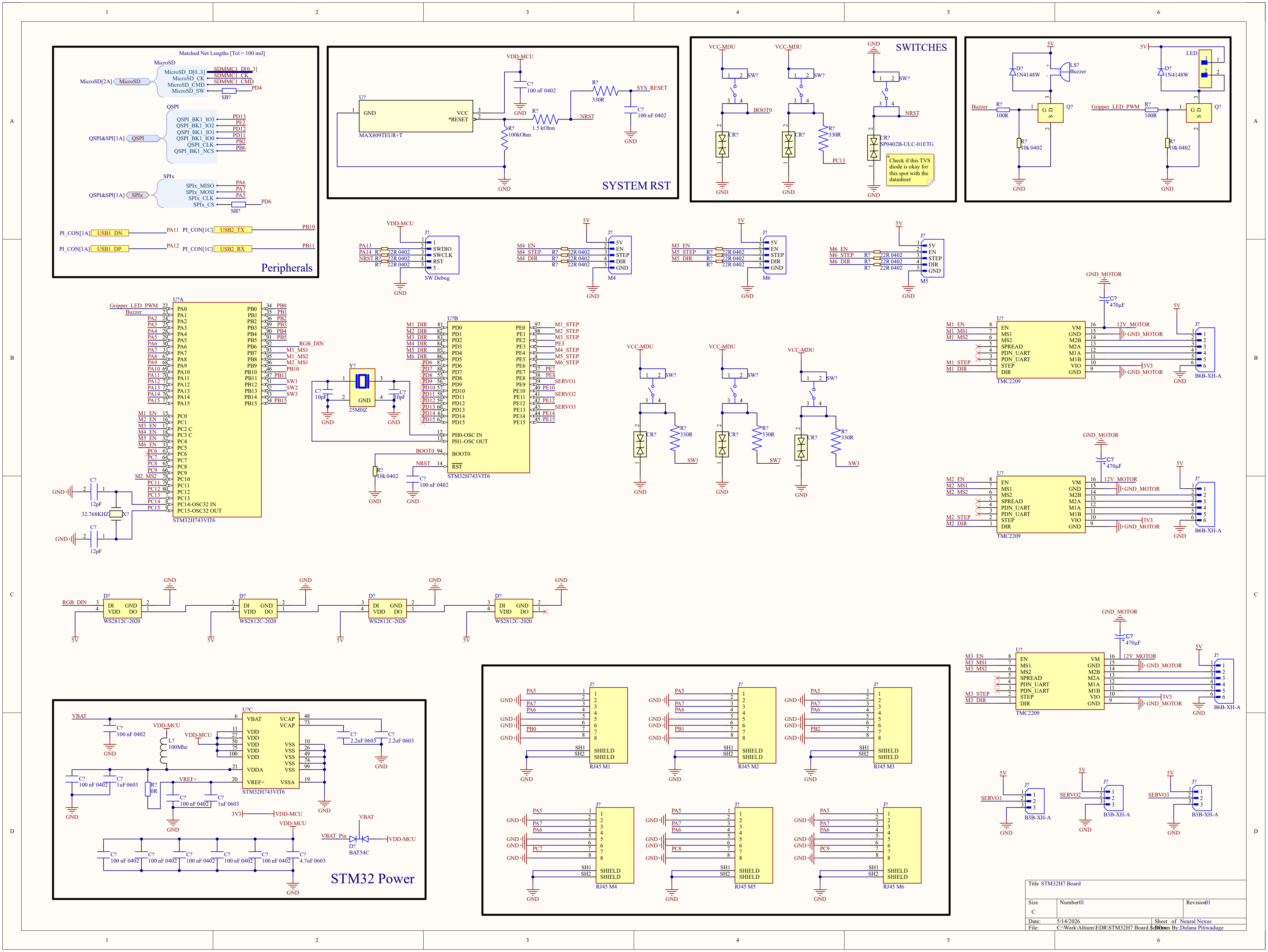
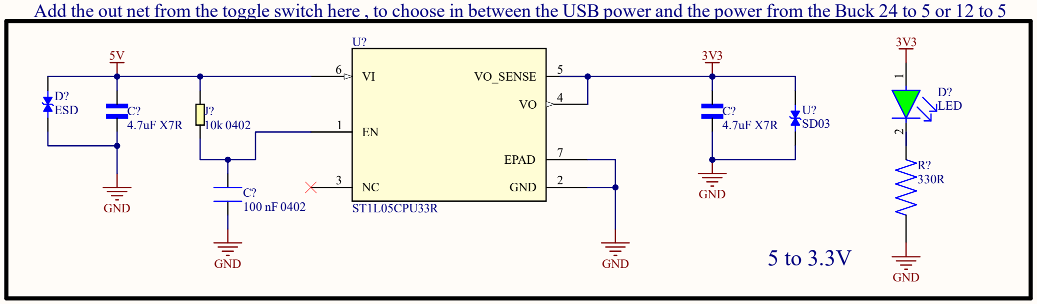


Peripherals
MicroSD
Matched Net Lengths [Tol = 100 mil]
QSPI
SPIx
SYSTEM RST
MAX809TEUR+T
SWITCHES
Check if this TVS diode is okay for this spot with the datasheet
SW Debug
STM32H743VIT6
25MHZ
32.768KHZ
TMC2209
B6B-XH-A
WS2812C-2020
STM32 Power
BAT54C
RJ45 M1
RJ45 M2
RJ45 M3
RJ45 M4
RJ45 M5
RJ45 M6
B3B-XH-A
Title STM32H7 Board
Size C
Number 01
Revision 01
Date: 5/14/2026
Sheet of Neural Nexus
File: C:\Work\Altium\EDR\STM32H7 Board.SchDoc
Drawn By: Dulana Pitiwaduge



Splitting the 12V line is only half the battle; you must also manage the ground return paths.

When those three motors pull 6A, that 6A has to flow back to the battery through the ground plane. If that 6A flows underneath or through the ground copper near your STM32 or your TPS56637 feedback resistors, it will raise the local ground voltage, causing your microcontroller to read false button presses, corrupt SPI data, or trigger a brown-out reset.

How to route it in Altium:

Do not use one massive, uniform ground pour for the whole board.

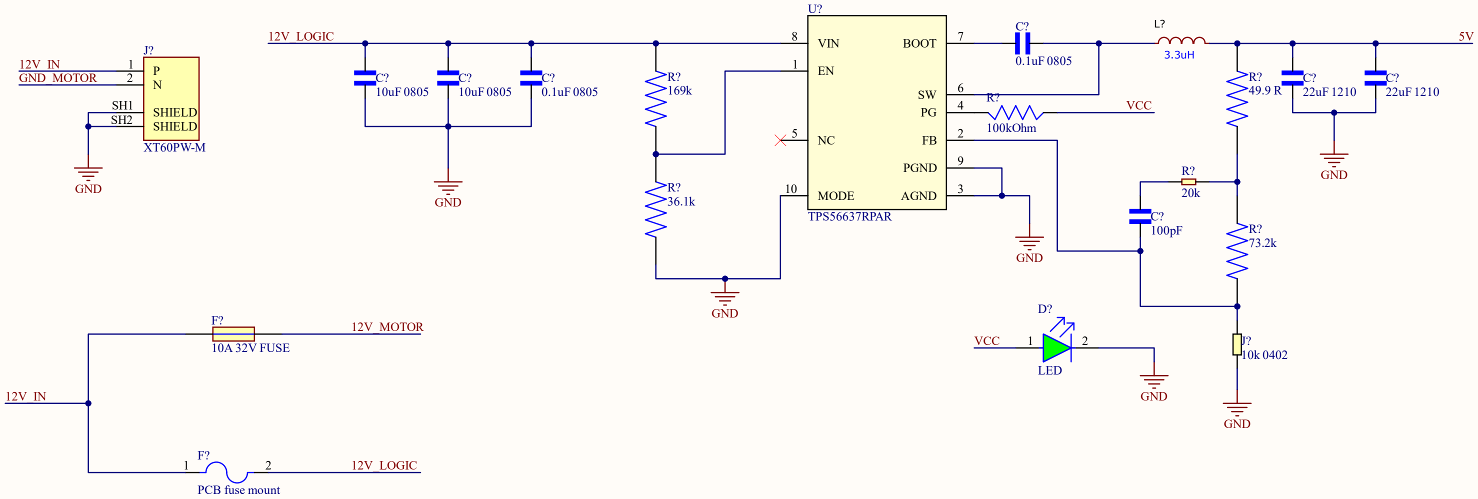
Create a specific, thick ground trace (or dedicated polygon pour) that connects the GND pins of the three TMC2209s directly back to the PGND pin of the XT60 connector.

Keep this motor ground return path physically away from your STM32 and analog sensors.

Your logic ground (PGND and AGND from the buck converter, and the STM32 ground) should meet the motor ground at exactly one single point: the physical solder pad of the XT60's negative pin.

3. Don't Forget the Bulk Capacitors!

As a final reminder, because you are running 3 motors from this 12V line, you absolutely must place a large aluminum electrolytic capacitor (e.g., 220uF to 470uF, rated for 35V or 50V) on the 12V_MOTORS net right next to the TMC2209 drivers. Without it, the inductive flyback from three simultaneous 2A motors will easily spike the 12V rail past 30V and detonate your chips.



Title POWER			
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Date: 5/14/2026		Sheet of	
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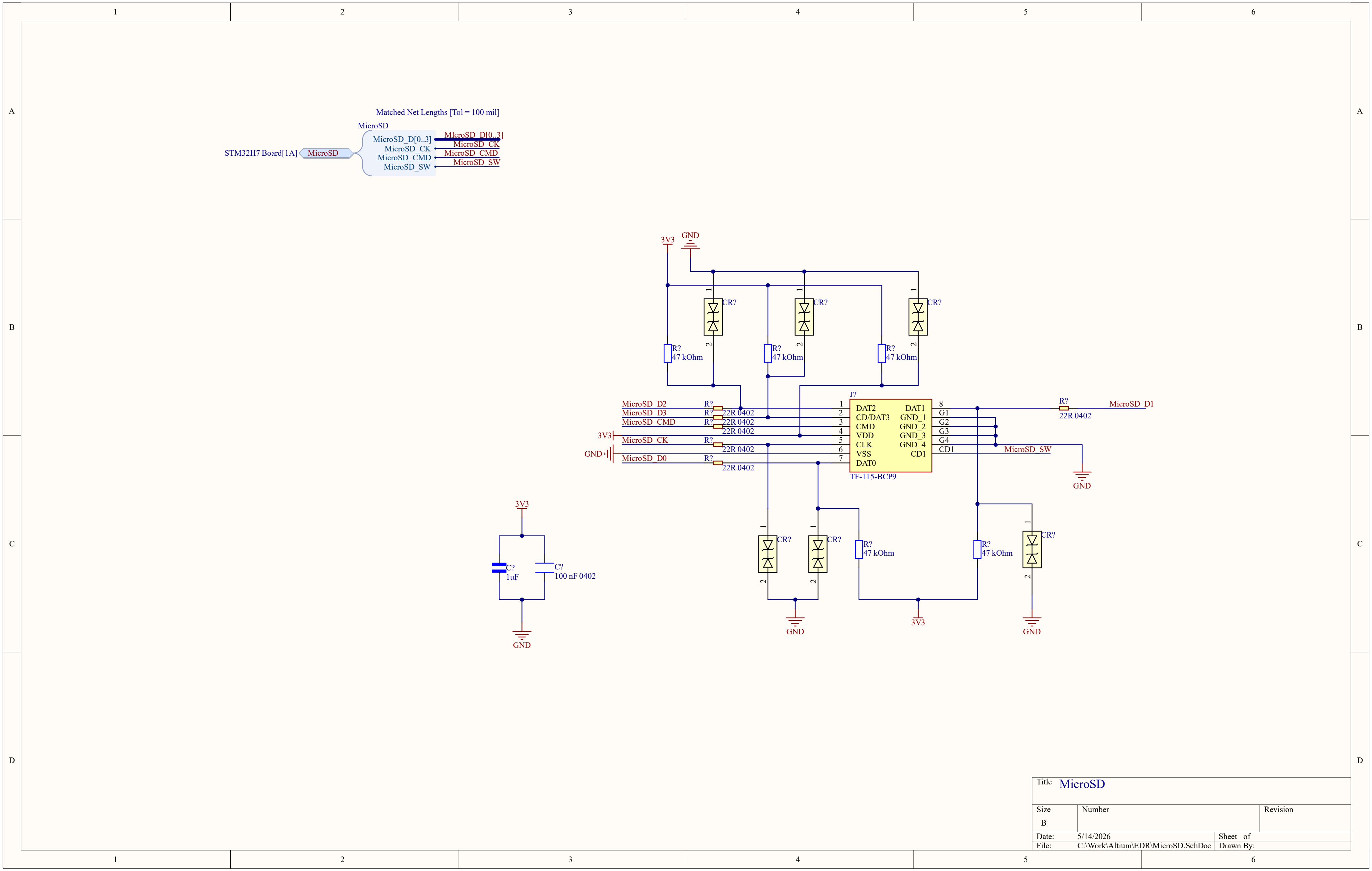
Matched Net Lengths [Tol = 100 mil]
MicroSD
STM32H7 Board[1A]
MicroSD
MicroSD_D[0..3]
MicroSD_CK
MicroSD_CMD
MicroSD_SW
MIcroSD_D[0..3]
MicroSD_CK
MicroSD_CMD
MicroSD_SW
3V3
GND
CR?
R?
47 kOhm
MicroSD_D2
MicroSD_D3
MicroSD_CMD
MicroSD_CK
MicroSD_D0
22R 0402
J?
DAT2
CD/DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
GND_1
GND_2
GND_3
GND_4
CD1
G1
G2
G3
G4
TF-115-BCP9
MicroSD_D1
MicroSD_SW
C?
1uF
100 nF 0402
Title MicroSD
Size B
Number
Revision
Date: 5/14/2026
File: C:\Work\Altium\EDR\MicroSD.SchDoc
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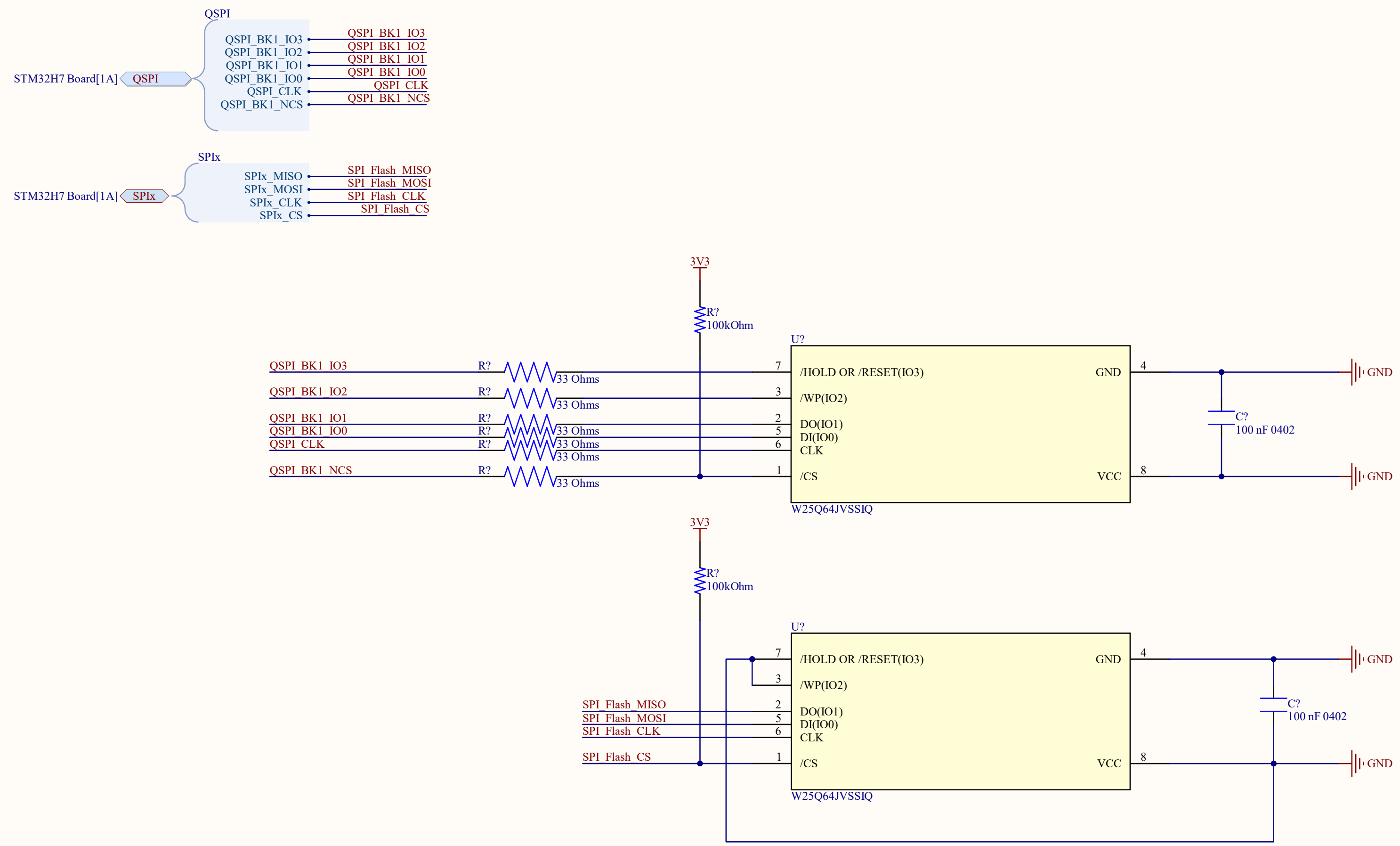
Drawn By:



Title	QSPI&SPI FLASH
Size	B
Number	
Revision	
Date:	5/14/2026
Sheet of	
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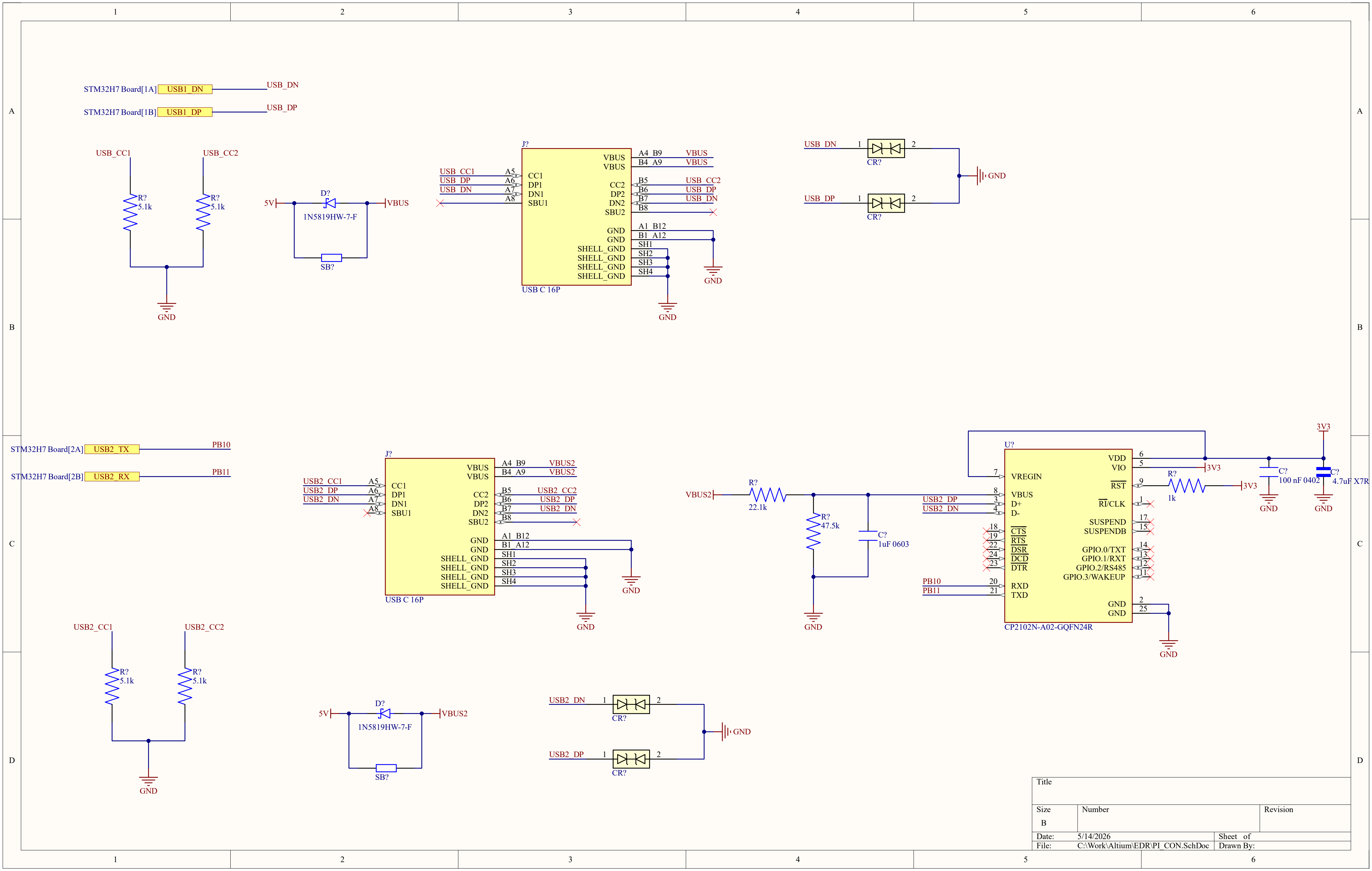
Drawn By:	


STM32H7 Board[1A] USB1_DN USB_DN
STM32H7 Board[1B] USB1_DP USB_DP
USB_CC1 USB_CC2
R? 5.1k
R? 5.1k
GND
5V D? 1N5819HW-7-F VBUS
SB?
J? USB C 16P
USB_CC1 A5 CC1
USB_DP A6 DP1
USB_DN A7 DN1
A8 SBU1
VBUS A4 B9 VBUS
VBUS B4 A9 VBUS
CC2 B5 USB_CC2
DP2 B6 USB_DP
DN2 B7 USB_DN
SBU2 B8
GND A1 B12
GND B1 A12
SHELL_GND SH1
SHELL_GND SH2
SHELL_GND SH3
SHELL_GND SH4
GND
USB_DN 1 CR? 2
USB_DP 1 CR? 2
GND
STM32H7 Board[2A] USB2_TX PB10
STM32H7 Board[2B] USB2_RX PB11
J? USB C 16P
USB2_CC1 A5 CC1
USB2_DP A6 DP1
USB2_DN A7 DN1
A8 SBU1
VBUS A4 B9 VBUS2
VBUS B4 A9 VBUS2
CC2 B5 USB2_CC2
DP2 B6 USB2_DP
DN2 B7 USB2_DN
SBU2 B8
GND A1 B12
GND B1 A12
SHELL_GND SH1
SHELL_GND SH2
SHELL_GND SH3
SHELL_GND SH4
GND
VBUS2 R? 22.1k
R? 47.5k
C? 1uF 0603
GND
U? CP2102N-A02-GQFN24R
7 VREGIN
8 VBUS
USB2_DP 3 D+
USB2_DN 4 D-
18 CTS
19 RTS
22 DSR
24 DCD
23 DTR
PB10 20 RXD
PB11 21 TXD
VDD 6
VIO 5
RST 9
RI/CLK 1
SUSPEND 17
SUSPENDB 15
GPIO.0/TXT 14
GPIO.1/RXT 13
GPIO.2/RS485 12
GPIO.3/WAKEUP 11
GND 2
GND 25
3V3
R? 1k 3V3
C? 100 nF 0402
C? 4.7uF X7R
3V3
GND
USB2_CC1 USB2_CC2
R? 5.1k
R? 5.1k
GND
5V D? 1N5819HW-7-F VBUS2
SB?
USB2_DN 1 CR? 2
USB2_DP 1 CR? 2
GND
Title
Size B
Number
Revision
Date: 5/14/2026
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File: C:\Work\Altium\EDR\PI_CON.SchDoc
Drawn By: